



AMAZON REVIEWS SENTIMENT ANALYSIS

Machine learning model

Abstract

This project develops a model to analyze Amazon reviews, identifying sentiment toward specific product aspects like “value for money” and “durability.” By focusing on individual attributes, it provides Amazon with targeted insights to enhance products and marketing. Key challenges include diverse review language, imbalanced aspect mentions, and mixed sentiments within reviews. Using data preprocessing and machine learning models, the project seeks to accurately capture and analyze aspect-specific sentiment, aiding Amazon in better understanding customer feedback.

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Project Proposal

Members:

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1. Introduction:

What is the project about? Why is it interesting? What makes it challenging? How do you plan to complete it?

This project focuses on creating a model that predicts how customers feel about specific parts of Amazon products based on their reviews. For example, it will analyze how customers feel about aspects like “value for money”, “durability”, “ease of use” etc. Understanding how customers feel can help business improve their products and marketing strategies. Analyzing feelings about specific aspects gives deeper insights compared to just knowing if the overall sentiment is positive, negative, or neutral. Additionally, as a regular Amazon customer, we are curious about how reviews play a crucial role in customer’s decisions to buy products. Working for Amazon ourselves, we recognize the importance of customer feedback and want to explore the impact of reviews on purchasing behavior.

Challenges:

1. It can be tough to pick out specific parts of a product mentioned in reviews since people write in different ways.
2. Some aspects might have fewer reviews than others, which could make it hard for the model to learn effectively.
3. Reviews can have mixed feelings, meaning the model needs to tell the difference between positive and negative sentiments within the same review.

Completion Plan:

1. Gather data from available Amazon review datasets, download the datasets and understand its structures such as features, types of data etc.
2. Clean up the data and figure out the aspects to analyze (remove duplicates, handle missing values, lowercasing, removing unwanted punctuation).
3. Label data, if the datasets do not have sentiment labels for specific aspects, we may need to label a subset of the data manually.
4. Choose right tools and libraries like Python programming language, Pandas, Scikit-learn and NumPy for data manipulation.
5. Build models (simple model like logistic regression and then move to more complex models), improve them to analyze sentiments about specific aspects.
6. Evaluate how well the models perform using the accuracy metrics.
 - a. Adjust models and techniques based on the results to improve accuracy, try different algorithms, improve data preprocessing techniques.

- b. Keep documentation of the project, this will help track progress and troubleshoot any issues you encounter.

2. Datasets:

Are there any dataset(s) you will use in your project? Provide basic statics about the datasets (e.g., number of samples, types of features, etc.). If you will curate your own dataset, highlight why you need to curate your own dataset, the process (rough) of curation, and other details, if any.

We plan to use Amazon product reviews dataset from Kaggle or Amazon's official dataset. As of now, we are unsure of data size (it depends on what subset we choose)

Types of features are:

- Review text
- Star rating (from 1 to 5)
- Product category
- Date of review (maybe)

If the dataset we choose does not cover specific aspects, we might need to create our own dataset by collecting more reviews through Amazon's API, and also by manually labelling some reviews to show specific aspects.

3. Evaluation metrics:

What quantitative metrics you will use to evaluate your approach?

Quantitative Metrics:

- Accuracy: The overall correctness of the sentiment predictions.
- Precision, Recall, and F1 Score: These metrics will help evaluate how well the model predicts feelings for each aspect.
- Confusion Matrix: A visual tool to see how well the model is performing and where it makes mistakes.

4. Baseline model:

What baseline models (e.g., k-NN, decision trees, linear models, SVM) you will use to compare with your approach.

In our sentiment analysis project, we have decided to use Logistic Regression as our baseline model. Logistic Regression is a widely used statistical method for binary classification tasks, making it well-suited for predicting the sentiment of Amazon product reviews.

Relevance to Our Project:

Since our goal is to classify customer reviews based on their sentiment regarding specific aspects of products (such as durability, ease of use, and value for money), Logistic Regression offers a straightforward and interpretable approach. It will help us establish a performance benchmark against which we can compare more complex models later.

5. Transition to Multi-Layer Perceptron's (MLPs):

Any idea of what kind of models (MLP/RNNs/CNNs/Transformers) you plan to use and why do you think your suggested models will be appropriate?

After evaluating the performance of Logistic Regression, we plan to enhance our model by in Transition to Multi-Layer Perceptron's (MLPs) incorporating Multi-Layer Perceptron's (MLPs). MLPs are a type of neural network that consists of multiple layers, allowing for more complex representations of the data.

Why MLPs are Appropriate:

- **Complexity Handling:** MLPs can capture non-linear relationships in the data, making them better suited for text data where the sentiment can be influenced by the interplay of multiple words and phrases.
- **Feature Learning:** MLPs automatically learn features from the input data, which can improve the model's ability to identify patterns associated with specific sentiments in reviews.
- **Improved Performance:** By moving to MLPs, we aim to achieve better accuracy and robustness in our sentiment predictions, especially as we explore more complex relationships within the review text.